

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (Credit System) Degree Examinations**

May - June 2023

19ME8X65 – AUTOMOTIVE ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw valve timing diagram for a 4-stroke petrol engine clearly indicating the position of following and explain them. (i) IVO (ii) IVC (iii) EVO (iv) EVC (v) Ignition	6	L*1	1	1
b)	Compare S.I. and C.I. engines.	4	L1	1	1
c)	Draw a neat sketch of Carter carburetor and explain how it satisfies different operating conditions of engine.	10	L2	2	1
2. a)	With neat sketches explain the construction and purpose of dry and wet cylinder liners.	7	L2	1	1
b)	Differentiate between air-cooled engine and water-cooled engine.	3	L1	1	1
c)	With a neat sketch explain the construction and working of S.U. Electrical fuel pump. What are the advantages and disadvantages of this pump over the A.C. Mechanical pump.	10	L2	2	1
3. a)	What is piston slap? Explain any two important methods to prevent piston slap with sketch.	6	L1	1	1
b)	With a neat sketch explain the working of overhead valve mechanism.	4	L2	1	1
c)	What are the factors on which the ignition advance requirements of an engine depend?	3	L1	2	1
d)	Describe the working principle of Bendix drive as a starting device with a neat sketch.	7	L2	2	1
Unit – II					
4. a)	With the help of a neat sketch, explain the working of a 3-speed sliding mesh gear box.	10	L2	3	1
b)	Explain the following types of rear axle supporting with sketch (i) semi floating axle (ii) full floating axle.	6	L2	4	1
c)	Differentiate between Hotchkiss and torque tube drive.	4	L2	4	1
5. a)	With a neat sketch explain the working of an overdrive mechanism.	6	L2	3	1
b)	With a neat sketch explain the working of fluid coupling.	4	L2	3	1
c)	With a neat sketch explain the working of Air suspension system in a vehicle.	7	L2	4	1
d)	Define the following (i) king pin inclination (ii) toe-in and toe-out.	3	L1	4	1
6. a)	Explain with a neat sketch the working of a single plate friction clutch.	7	L2	3	1
b)	What are the requirements of a good clutch?	3	L1	3	1
c)	With a neat sketch explain the working of differential of an automobile.	10	L2	4	1

P.T.O.

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Draw the layout of a hydraulic braking system and show the arrangement of various components. | 10 | L2 | 5 | 1 |
| b) | Describe the merits and demerits of hydraulic braking system. | 3 | L1 | 5 | 1 |
| c) | Describe the working of vacuum servo braking system with a neat sketch. | 7 | L2 | 5 | 1 |
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| 8. a) | Write a short note on following: | | | | |
| | i) Tubed and tubeless tyre. | 10 | L2 | 5 | 1 |
| | ii) Exhaust Gas Recirculation. | 7 | L1 | 5 | 1 |
| b) | Explain the working of Drum type brake with a neat sketch. | 7 | L1 | 5 | 1 |
| c) | Enumerate the advantages of Disc brakes over Drum brakes. | 3 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
